Semiconductor Manufacturing International Corporation

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Document Level: (For Engineering & Quality Document/工程暨品质文件专用)

☐ Level 1 - Manual ☑ Level 2 – Procedure/SPEC/Report ☐ Level 3 - Operation Instruction

Security Level:

☑ Security 1 - SMIC Confidential ☐ Security 2 - SMIC Restricted ☐ Security 3 - SMIC Internal

Document Change History

Doc. Rev.	Tech Dev. Rev.	Effective Date	Author	Change Description
0	1.14_0	2016-06-14	Jason Sun	Initial release on DMS. Please be kindly noted that the contents of this document may contain the confidential information of a third party such as SMIC's technology licensor, IP vendor, or customer, etc. Pursuant to the agreement between such third party and SMIC, SMIC's employee with a need to access such third party's confidential information shall (i) use such information solely for the benefit of SMIC or the third party, and for no other purpose, (ii) keep such information confidential for a period of fifteen years or longer from its disclosure of such information to SMIC, and (iii) agree that the third party is a third party beneficiary for the confidentiality obligation stated herein.

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1 Title

40nm Logic Low Leakage Salicide 1.1V/1.8V/2.5V 1P9M with 2TMc and ALPA2 StarRC Runsets for CCI flow.

2 Purpose

This PEX document is intended to be a release note for the users who adopt SMIC 40nm LL technology and apply StarRC for parasitic extraction.

3 Scope

This document provides the related information about StarRC runsets for 40nm LL technology.

4 Nomenclature

LVS runset	**L**ayout **V**ersus **S**chematic runset
PEX runset	**P**arasitic **EX**traction runset
CCI	**C**alibre **C**onnectivity **I**nterface

5 Reference

This document and the corresponding PEX runsets are created based on the following documents,

5.1 Design Rule

40nm Logic Salicide 0.9/1.1/1.2/1.8/2.5V low leakage and generic Design Rule
Doc. No. TD-LO40-DR-2001 (Tech Version:1.10 Revision:3)

5.2 Process Control Monitor (PCM) Specification

40nm Logic Low Leakage (LL) 1P10M 1.1/2.5V PCM SPEC
Doc. No. TD-LO40-PS-2001 (Tech Version:1.16 Revision:7)

5.3 SPICE Model

40nm Logic Low Leakage 1P10M(1P9M,1P8M,1P7M,1P6M) Salicide 1.1V/1.8V/2.5V SPICE Model (Version 1.4_2R)
Doc. No. TD-LO40-SP-2001 (Tech Version:1.14 Revision:2)

5.4 LVS runset

40nm Logic Low Leakage Salicide 1.1V/1.8V/2.5V Calibre LVS for StarRC
Doc. No. TD-LO40-LC-2003 (Tech Version:1.14_0 Revision:0)

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6 Responsibility:

Technology R&D\ETD\Tech-Bas Des-Enb\Compact Model 2 \Post-layout Extraction team

7 Subject Content:

7.1 Capabilities

This section summarizes the criteria of the related product versions for 40nm Logic Low Leakage StarRC runsets for CCI flow as follows,

7.1.1 Platform

StarRC version 2012.06-SP1 or later.

7.1.2 Connection Products

Calibre LVS version 2013.1_14.11 or later

7.2 Contents

7.2.1 The Compressed File Structures

Every compressed TAR file of StarRC package contains the following files,

(1) ***hcell_list*** The device list for Calibre hierarchical and hybrid extraction, mainly for RF devices. It will preserve RF cells as a hierarchical cells so that StarRC could treat them as gray boxes during parasitic extraction.
(2) ***query_cmd*** Calibre commend file for Calibre CCI flow
(3) ***CCI_cmd*** StarRC commend file for Calibre CCI flow.
(4) ***starrc*_CCI.map*** StarRC mapping file for Calibre CCI flow.
(5) ***starrc*_cell.map*** Cell level mappping for Astro techfile.
(6) ***.itf*** ITF which is specify the cross-sectional profile of BEOL.
(7) ***.nxtgrd*** NXTGRD database for parasitic extraction.
(8) ***ReleaseNote*** specifies the revision history and fundamental information of StarRC runsets.

With regard to the usage of these files, please refer to the attached files:

- ***Quick-start on SMIC Calibre Connectivity Interface – StarRC flow***

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Fig. 01 shows the detail file structures of StarRC compressed TAR package for CCI_StarRC_40LL_1P9M_8Ic_1TMc_ALPA1 as an example. The file structures are similar for the other metallization options of StarRC runsets.

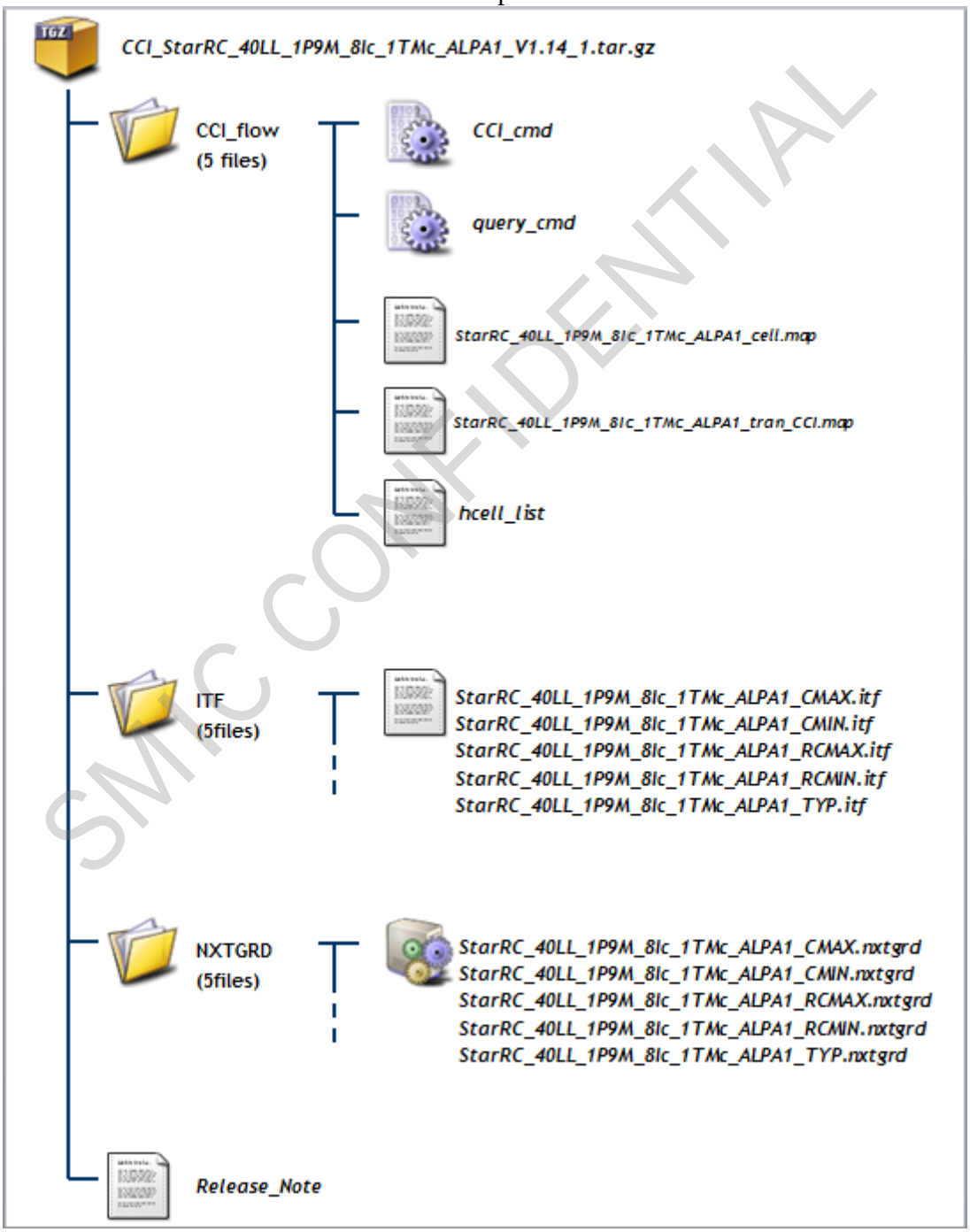

Fig. 01 The file structure of StarRC compressed TAR package

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7.2.2 Metallization Options

The following table summarizes the available metallization options included in this runset of 40LL technology. The options denoted as **black-bold** are included in this document.

The Number of Metal Layers		4		5				6				7				8				9				10	
Metallization Option		1P4M_1TM_ALPA2	1P4M_MTT_ALPA2	1P5M_1TM_ALPA2	1P5M_2TM_ALPA2	1P5M_MTT_ALPA2	1P5M_1TM_MTT_ALPA2	1P6M_1TM_ALPA2	1P6M_2TM_ALPA2	1P6M_MTT_ALPA2	1P6M_1TM_MTT_ALPA2	1P7M_1TM_ALPA2	1P7M_2TM_ALPA2	1P7M_MTT_ALPA2	1P7M_1TM_MTT_ALPA2	1P8M_1TM_ALPA2	1P8M_2TM_ALPA2	1P8M_MTT_ALPA2	1P8M_1TM_MTT_ALPA2	1P9M_1TM_ALPA2	**1P9M_2TM_ALPA2**	1P9M_MTT_ALPA2	1P9M_1TM_MTT_ALPA2	1P10M_2TM_ALPA2	1P10M_1TM_MTT_ALPA2
M1	**M1**	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	**X**	X	X	X	X
	V1	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	**X**	X	X	X	X
Mn (n = 2 ~ 8)	**M2**	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	**X**	X	X	X	X
	V2	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	**X**	X	X	X	X
	M3	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	**X**	X	X	X	X
	V3			X		X		X	X	X	X	X	X	X	X	X	X	X	X	X	**X**	X	X	X	X
	M4			X		X		X	X	X	X	X	X	X	X	X	X	X	X	X	**X**	X	X	X	X
	V4							X		X		X	X	X	X	X	X	X	X	X	**X**	X	X	X	X
	M5							X		X		X	X	X	X	X	X	X	X	X	**X**	X	X	X	X
	V5											X		X		X	X	X	X	X	**X**	X	X	X	X
	M6											X		X		X	X	X	X	X	**X**	X	X	X	X
	V6															X		X		X	**X**	X	X	X	X
	M7															X		X		X	**X**	X	X	X	X
	V7																			X		X		X	X
	M8																			X		X		X	X
Top Metal (9k)	**TV1**				X		X		X		X		X		X		X		X		**X**		X	X	X
	TM1				X		X		X		X		X		X		X		X		**X**		X	X	X
	TV2	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	**X**	X	X	X	X
	TM2	X		X	X			X	X			X	X			X	X			X	**X**			X	
Thick Top Metal (34k)	**MTT**		X			X	X			X	X			X	X			X	X			X	X		X
AL RDL (2)	**BCB1**	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	**X**	X	X	X	X
	RDL	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	**X**	X	X	X	X

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7.2.3 Corners

Based on in-line process and PCM specification, we provide the following 4 corners of post-layout extraction and we recommend users to apply these runsets for design reference,

Cmin this corner is intended to provide the best-case of interconnect capacitance in post-layout extraction as Fig. 02. The physical dimension of conduction layers and dielectric layers will be skewed to the lower limit and the upper limit, respectively. It will result in the lowest coupling capacitances but the higher line resistances in post-layout extraction. The resistances of contact and via are kept the same as typical runset.

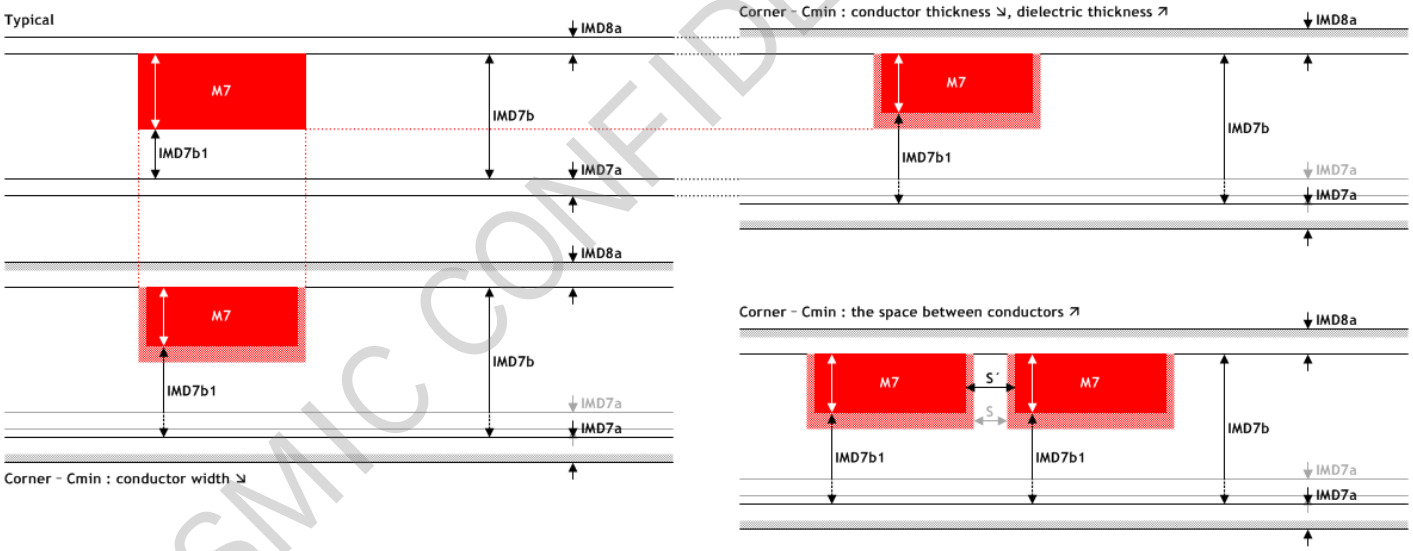

Fig. 02 The comparison of **Typical** cross section with **Cmin** corner

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Cmax this corner is intended to provide the worst-case of interconnect capacitance in post-layout extraction as Fig. 03. The physical dimension of conduction layers and dielectric layers will be skewed to the upper limit and the lower limit, respectively. It will result in the highest coupling capacitances but the lower line resistances in post-layout extraction. The resistances of contact and via are kept the same as typical runset.

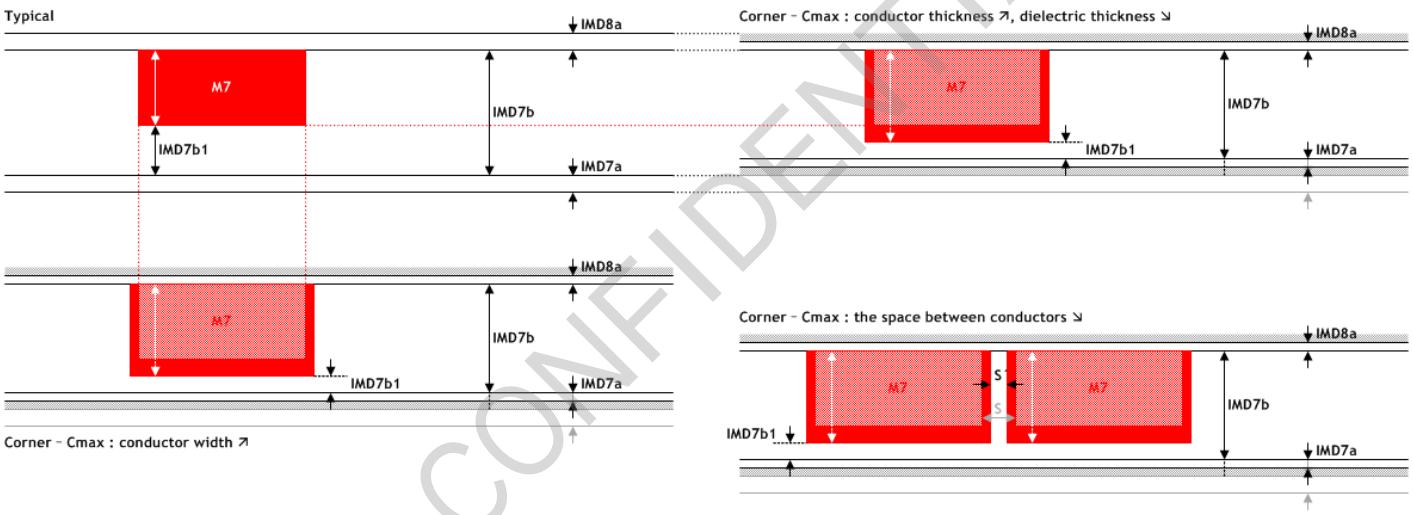

Fig. 03 The comparison of **Typical** cross section with **Cmax** corner

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RCmin this corner is intended to provide the best-case in post-layout extraction by compromising the interconnect resistance and capacitance as Fig. 03. The physical dimension of conduction layers and dielectric layers will be both skewed to the upper limit. It will result in not only the lowest line resistances but also the relatively lower capacitances in post-layout extraction. The resistances of contact and via are also skewed to the lower limit of PCM specification.

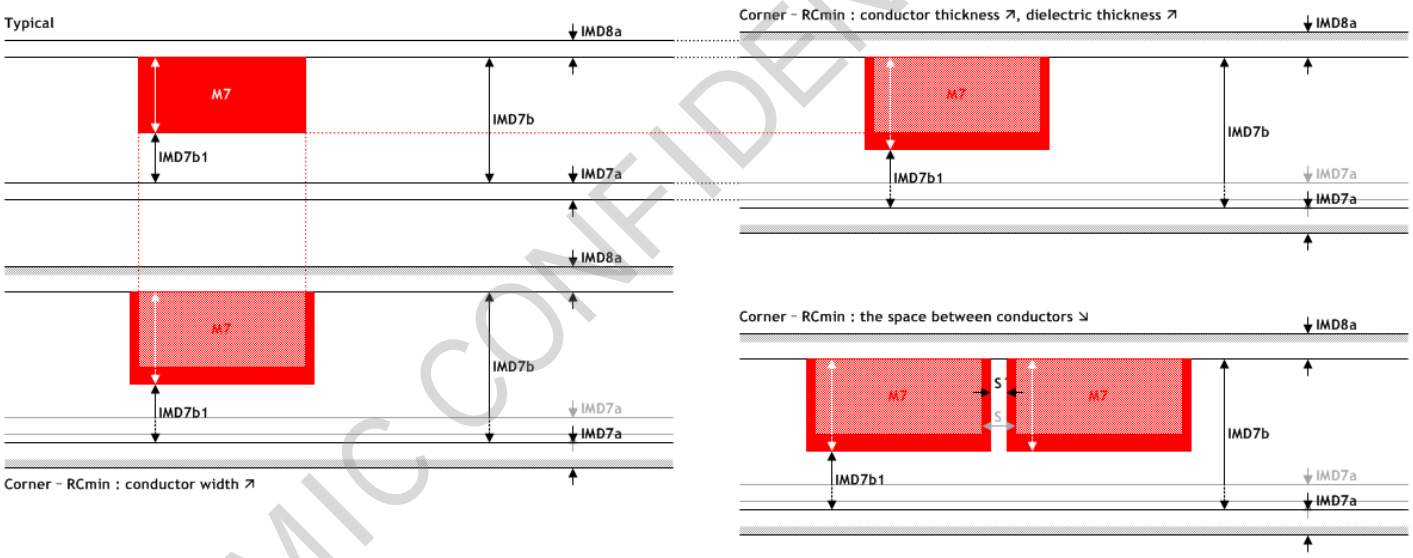

Fig. 04 The comparison of **Typical** cross section with **RCmin** corner

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RCmax this corner is intended to provide the worst-case in post-layout extraction by compromising the interconnect resistance and capacitance as Fig. 04. The physical dimension of conduction layers and dielectric layers will be both skewed to the lower limit. It will result in not only the highest line resistances but also the relatively higher capacitances in post-layout extraction. The resistances of contact and via are also skewed to the upper limit of PCM specification.

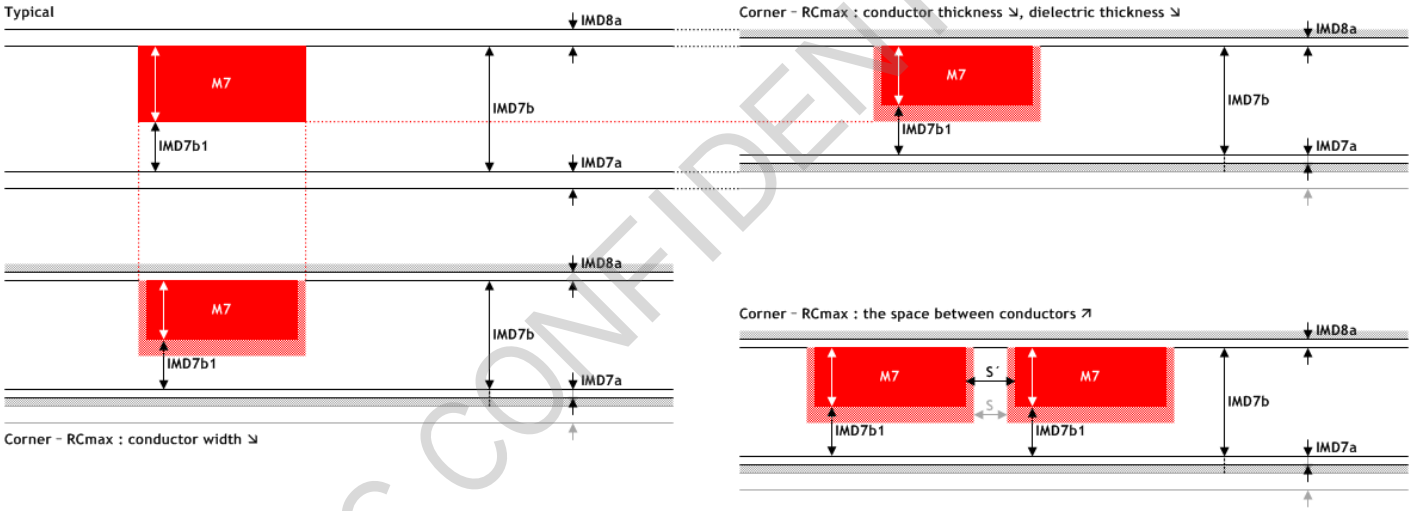

Fig. 05 The comparison of **Typical** cross section with **RCmax** corner

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7.3 Know Issue

7.3.1 Calibre CCI – StarRC Flow

The following warnings may appear during extractions but they could be ignored,

[WARNING I] This is due to lack of RPV and AREA setting in "via_layers" in mapping file
WARNING:(5301) StarXtract
WARNING: Setting RPV*AREA value of 1e-6 for ntd_nwell_subcont
WARNING:(5301) StarXtract
WARNING: Setting RPV*AREA value of 1e-6 for ptd_pwell_subcont
WARNING:(5301) StarXtract
WARNING: Setting RPV*AREA value of 1e-6 for ptd_rwell_subcont
WARNING:(5301) StarXtract
WARNING: Setting RPV*AREA value of 1e-6 for ptd_dnwd_psub_subcont
WARNING:(5301) StarXtract
WARNING: Setting RPV*AREA value of 1e-6 for ntd_dnwell_subcont

[WARNING II] This is due to syntax discrepancy between Calibre LVS deck and StarRC runset
WARNING: StarXtract
WARNING: CALIBRE_RUNSET contains unsupported pre-processor directive "!TVF". Supported directives
WARNING: are DEFINE UNDEFINE IFDEF IFNDEF ELSE and ENDIF. The following directive will be ignored;
WARNING:
SmicSP2R_cal40_LL_sali_p1mtx_11182533_V1.14_0.lvs_for_cci_new@1: #!tvf
WARNING: StarXtract
WARNING: Skip encrypted lines from line @16501:
WARNING:
WARNING: StarXtract
WARNING: Runset via layer ppolycont connects down to MOM_poly[npoly] that has a different picture
WARNING: level than that of via's GRD from-layer ppoly

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8 Attachment

8.1 Usage Guide

quick-start_on_smic_cci-starrc.pdf (PDF format)

8.2 PEX Compressed TAR Files

```
CCI_StarRC_40LL_1P9M_7Ic_2TMc_ALPA2_V1.14_0.tar.gz
CCI_StarRC_40LL_1P9M_7Ic_2TMc_ALPA2_V1.14_0
|-- CCI_flow
|     |-- CCI_cmd
|     |-- StarRC_40LL_1P9M_7Ic_2TMc_ALPA2_cell.map
|     |-- StarRC_40LL_1P9M_7Ic_2TMc_ALPA2_tran_CCI.map
|     |-- hcell_list
|     `-- query_cmd
|-- ITF
|     |-- StarRC_40LL_1P9M_7Ic_2TMc_ALPA2_CMAX.itf
|     |-- StarRC_40LL_1P9M_7Ic_2TMc_ALPA2_CMIN.itf
|     |-- StarRC_40LL_1P9M_7Ic_2TMc_ALPA2_RCMAX.itf
|     |-- StarRC_40LL_1P9M_7Ic_2TMc_ALPA2_RCMIN.itf
|     `-- StarRC_40LL_1P9M_7Ic_2TMc_ALPA2_TYP.itf
|-- NXTGRD
|     |-- StarRC_40LL_1P9M_7Ic_2TMc_ALPA2_CMAX.nxtgrd
|     |-- StarRC_40LL_1P9M_7Ic_2TMc_ALPA2_CMIN.nxtgrd
|     |-- StarRC_40LL_1P9M_7Ic_2TMc_ALPA2_RCMAX.nxtgrd
|     |-- StarRC_40LL_1P9M_7Ic_2TMc_ALPA2_RCMIN.nxtgrd
|     `-- StarRC_40LL_1P9M_7Ic_2TMc_ALPA2_TYP.nxtgrd
`-- Release_Note
```

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